

Question	Answer	Marks
7(a)	energy (transferred) per (unit) charge	B1
7(b)	$V = 0.25 \times 6$ $= 1.5$	C1
	$Ir = E - IR$ $Ir = 1.8 - 1.5$ $= 0.3$	C1
	$r = 0.3 / 0.25$ $= 1.2 \Omega$	A1
	or (Total) $R = 1.8 / 0.25$ $= 7.2$ $E / I = (R + r)$	(C1)
	$1.8 / 0.25 = 6 + r$	(C1)
	$r = 7.2 - 6$ $= 1.2 \Omega$	(A1)
7(c)(i)	The same	B1
7(c)(ii)	Any 3 from: - before t_1 / when current constant, the (total) resistance is constant - at t_1 / when current increases, the (total) resistance decreases (due to decrease of external resistance) (after t_1) temperature (of lamp) increases (so the resistance of the lamp increases) (after t_1) resistance of lamp increases (so total resistance increases so the current in the ammeter decreases)	B3

9702/22

Cambridge International AS & A Level – Mark Scheme

February/March 2024

PUBLISHED

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